

Phase-3 Project Design

Date	28-09-2026
Team ID	SWTID-2026-7624
Project name	Script – controlled ACL
Maximum marks	4 marks

Here is the complete documentation for Phase 3: Project Design for your ServiceNow project, structured specifically for inclusion in your report or submission:

Phase 3: Project Design

1. Data Architecture & Schema Design

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The core storage structure for the application is built on a custom ServiceNow table designed to hold both student and departmental information.

A. Table Configuration

Table Label	Institution Details
Table Name	u_institution_details
Extends	None (Extends for = false)

The core storage structure for the application is built on a custom ServiceNow table designed to hold both student and departmental information.

A. Table Configuration

Table Label: Institution Details

Table Name: u_institution_details

Extends: None (Extends for = false)

B. Field Dictionary Schema

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Field Name	Column Name	Data Type	Reference / Details	Purpose
Student Roll Number	u_student_roll_number	Auto Number	System generated ID	Unique record identifier
Student Name	u_student_name	Reference	sys_user	Links to student user account
Faculty Name	u_faculty_name	Reference	sys_user	Links to assigned faculty member
Branch	u_branch	Choice	ECE, EEE, CSE	Categorizes record for ACL evaluation
Email	u_email	String	Standard text	Student contact email
Phone Number	u_phone_number	String	Standard text	Student contact phone
Description	u_description	String (Multi-Line)	Multi-line text block	Additional departmental notes

2. Security Architecture & Role Design

Access control is enforced using a combination of ServiceNow Custom Roles and Script-Controlled Access Control Lists (ACLs) to separate operational capabilities and isolate branch data.

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A. Role Mapping Matrix

Role Name	Access Type	Scope & Description
bb1	Read	Allows viewing records where Branch = EEE.
bb2	Create	Enables the "New" button to insert new records.
bb3	Write / Edit	Grants permission to modify existing record fields.
bb4	Delete	Grants permission to delete records from the table.
admin	Global / Full	Unrestricted access across all operations and branches.

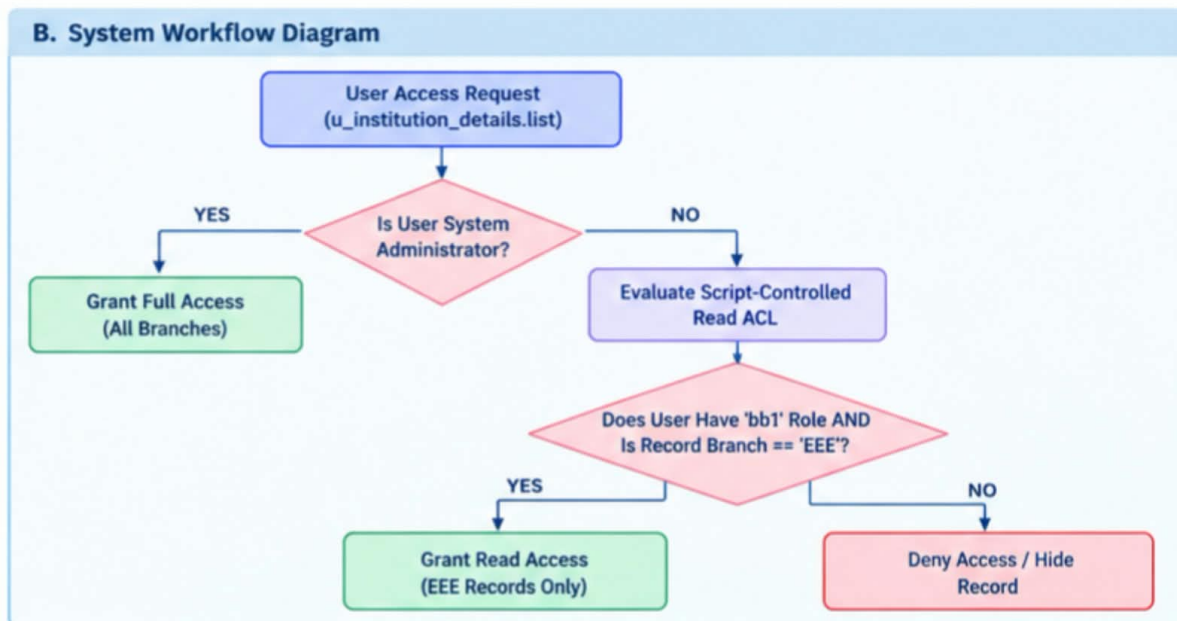
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B. System Workflow Diagram



3. Script-Controlled ACL Logic Design

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Read Access Control List (ACL) Design

Type	Record
Operation	read
Name	u_institution_details
Active	true
Advanced	true
Requires Role	bb1
Condition	Branch is EEE

Execution Script

```
(function () {  
    // Allow System Administrators full access  
    // to all records  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only users with the bb1 role to view  
    // EEE branch records  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
    // Deny access to all other users or branches  
    return false;  
})();
```

(Source: ServiceNow Access Control List Script Specification)

Read Access Control List (ACL) Design

Type: Record

Operation: read

Name: u_institution_details

Active: true

Advanced: true

Requires Role: bb1

Condition: Branch is EEE

Execution Script:

```
(function () { // Allow System Administrators full access to all records if (gs.hasRole('admin'))  
{ return true; } // Allow only users with the bb1 role to view EEE branch records if  
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(Source: ServiceNow Access Control List Script Specification)

4. Operations & CRUD Design

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To enforce proper separation of duties, operational ACLs are designed independently from the Read ACL:

Operation	Table	Requires Role	Logic
Create (create)	u_institution_details	bb2	Allows form submission for record creation without needing record-level conditions.
Write (write)	u_institution_details	bb3	Unlocks record fields for inline or form updating.
Delete (delete)	u_institution_details	bb4	Displays the "Delete" UI Action button on record forms and list choices.

To enforce proper separation of duties, operational ACLs are designed independently from the Read ACL:

Create ACL (create):

Table: u_institution_details

Requires Role: bb2

Logic: Allows form submission for record creation without needing record-level conditions.

Write ACL (write):

Table: u_institution_details

Requires Role: bb3

Logic: Unlocks record fields for inline or form updating.

Delete ACL (delete):

Table: u_institution_details

Requires Role: bb4

Logic: Displays the "Delete" UI Action button on record forms and list choices.